AS · Unit 2 · WCH12 Energetics, Group Chemistry, Halogenoalkanes and Alcohols

Introduction to kinetics and equilibria

Kinetics asks how fast a reaction goes; equilibrium asks how far it goes. This topic explains rates using collision theory and the Maxwell–Boltzmann distribution of molecular energies, shows how catalysts work, and then uses Le Chatelier's principle to predict how a reversible reaction responds when you change the conditions, including the compromises made in industry.

SPEC 9.1 Collision theory: concentration, temperature, pressure and surface area

WHAT THIS SPEC POINT ASKS

Use collision theory to explain how concentration, temperature, pressure and surface area change the rate of a reaction.

What has to happen for particles to react

- Particles must **collide**.
- The collision must have **energy at least equal to the activation energy** (E_a), so bonds can start to break (9.2).
- The particles must meet with the **right orientation**, so the reacting parts of the particles actually touch.
- A collision that meets these conditions is a **successful** (or effective) collision. Most collisions aren't successful: the particles just bounce apart.
- **Rate** depends on the **number of successful collisions per unit time** (the *frequency* of successful collisions), not just on how many collisions there are in total.

The four factors

Change	What happens to the particles	Effect on rate
Higher concentration (solutions)	more particles **in the same volume**, so they collide more often	more successful collisions **per unit time**
Higher pressure (gases only)	the same gas particles are squeezed into a smaller volume: more particles per unit volume	more successful collisions per unit time
Larger surface area (solids, e.g. powder	more particles are **exposed** at the surface and available to collide	more successful collisions per unit time

Change	What happens to the particles	Effect on rate
instead of lumps)		
Higher temperature	particles have more **kinetic energy**: (1) a much **greater proportion have energy ≥ E_a**; (2) they also move faster, so collide a little more often	many more successful collisions per unit time; effect (1) is by far the bigger one (see 9.4)

- Concentration, pressure and surface area change **how often** particles collide. They don't change the energy of the particles, so the **fraction** of collisions that succeed stays the same.
- Temperature is different: it changes the **energy** of the particles, so it changes the fraction of collisions that succeed. That's why a 10 °C rise can roughly double the rate of many reactions.
- Pressure only matters when at least one reactant is a **gas**. Changing the pressure on a reaction in solution has no real effect.

EXAM TIP

Build every "explain the effect on rate" answer from three links: **what changes about the particles → more successful collisions → per unit time** (or "more frequent"). Leaving out "per unit time" or "successful" is the most common way to drop a mark.

MARK-SCHEME WORDING

- Concentration: "more particles **in the same volume**" and "more (successful) collisions **per unit time**". ALLOW "particles are closer", "frequency of collisions". IGNORE "chance / probability of collisions". "More particles so more successful collisions" scores only 1 (WCH12 Jan 2025).
- Surface area: "larger surface area", ALLOW "more (active) sites". IGNORE "greater area of contact".
- Temperature: "particles have more kinetic energy", "**more particles have energy ≥ E_a**", "more successful collisions per unit time".
- Pressure: "more collisions per unit time". Any reference to **atoms** (instead of molecules or particles) can lose the mark.

Often asked as: explain why a change in concentration, pressure or catalyst particle size speeds up a reaction, usually inside an industrial-process question.

COMMON MISTAKE

"Particles collide more" with no time reference. Say **more frequent** collisions or **more collisions per second**.

"Higher concentration gives particles more energy." It doesn't: only temperature changes particle energy.

Mixing up the explanation for **rate** with the explanation for **yield**. Pressure can raise both, but for different reasons: collisions for rate, Le Chatelier for yield (9.10). In a two-mark answer the examiners want one consistent route.

SPEC 9.2 Activation energy

WHAT THIS SPEC POINT ASKS

Understand that a collision only leads to reaction if it has enough energy: at least the activation energy.

Definition

- **Activation energy (E_a)** is the **minimum energy** that colliding particles need for a reaction to happen.
- It is the energy needed to start breaking bonds in the reactants. On a reaction profile it's the height of the "hill" from the **reactants** up to the **peak** (the transition state).
- Every reaction has an E_a, whether it's exothermic or endothermic. Its size isn't set by ΔH: a very exothermic reaction can still have a large E_a (that's why fuels don't catch fire until you light them).
- For an **endothermic** reaction E_a must be at least as big as ΔH, because the products sit higher than the reactants.

Reading a reaction profile

- x-axis: **progress of reaction**; y-axis: **enthalpy** (or energy).
- E_a: arrow from the **reactant line up to the peak**.
- ΔH: arrow from the **reactant line to the product line** (points down for exothermic, up for endothermic).
- See the figure in 9.6 for a labelled example.

MARK-SCHEME WORDING

- "Particles with energy ≥ E_a" (ALLOW "collisions with energy greater than E_a").
- On a profile, E_a is drawn as a rise **from the reactants to the peak**. An arrow starting at the products, or ending below the peak, loses the mark (WCH12 Jan 2022).

Often asked as: label E_a on a profile (often for an endothermic reaction); use E_a inside a Maxwell–Boltzmann or catalyst explanation.

COMMON MISTAKE

Drawing E_a from the products to the peak (that's the E_a of the **reverse** reaction).

Calling E_a "the energy needed for the reaction": say **minimum** energy, and that it's needed for a **successful collision**.

SPEC 9.3 Calculating rate: 1/time and gradients

WHAT THIS SPEC POINT ASKS

Calculate a rate (i) as 1/time from the time taken for a reaction, and (ii) from the gradient of a graph, by drawing a tangent, for the initial rate or the rate at time t.

What rate means

- **Rate of reaction** = change in amount (or concentration) of a reactant or product **per unit time**.
- Units follow what you measured: mol dm^{-3} s^{-1} for concentration, cm^3 s^{-1} for a gas volume, g s^{-1} for a mass.

Ways to follow a reaction

Method	Use it when	Example
Gas volume (gas syringe or upturned measuring cylinder/burette over water)	a gas is made	marble chips + HCl; decomposition of H_2O_2
Mass loss (open flask on a balance)	a **dense** gas escapes (CO_2 works; H_2 is too light to measure well)	$CaCO_3$ + HCl
Time for a set change ("clock" method)	a colour appears or a precipitate hides a mark	sodium thiosulfate + HCl (cloudy sulfur hides a cross)
Sampling and titration	concentration changes but nothing is visible	take samples, **quench** them (stop the reaction, e.g. by cooling or diluting), titrate

(i) Rate = 1/time

- In a clock reaction you measure the **time** for a fixed amount of reaction (for example, the cross disappearing).
- A shorter time means a faster rate, so **rate** $\propto$ **1/time**. The value 1/t (units s^{-1}) is used as a measure of the rate.
- Plot 1/t against concentration (or temperature) to see how rate depends on it. A straight line through the origin means rate is proportional to concentration.

WORKED EXAMPLE

Worked example: rate from 1/time

Sodium thiosulfate reacts with dilute HCl: $S_2O_3^{2-}(aq) + 2H^+(aq) \rightarrow S(s) + SO_2(g) + H_2O(l)$. The sulfur makes the mixture cloudy.

The cross under the flask disappears after **40 s** with 0.10 mol dm^{-3} thiosulfate, and after **80 s** with 0.05 mol dm^{-3}.

Rate (0.10 mol dm^{-3}) = 1/40 = **0.025 s^{-1}** (2.5 × 10^{-2} s^{-1})

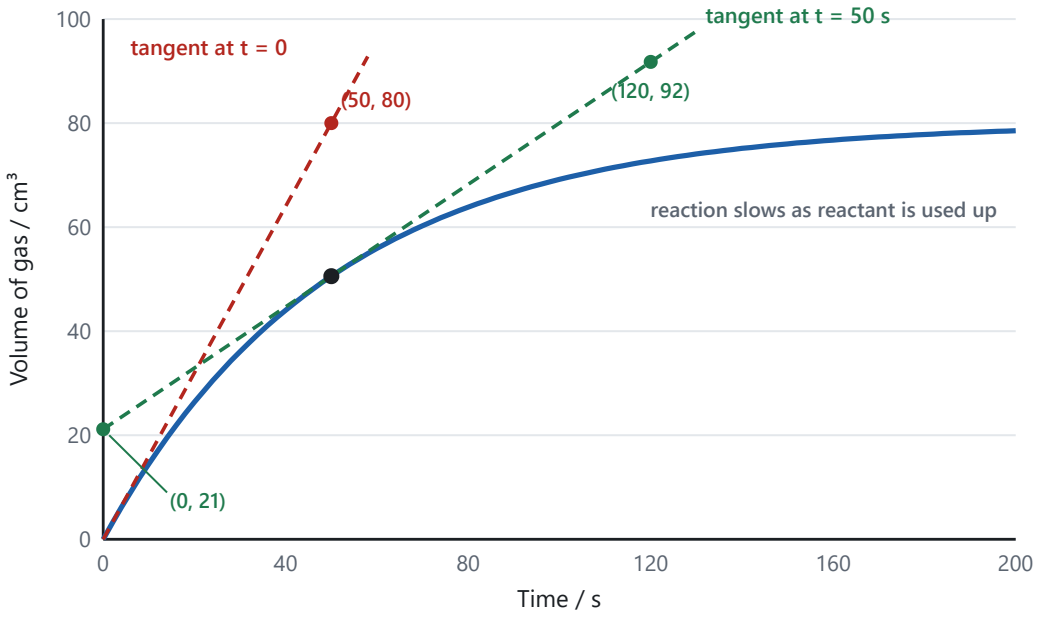
Rate (0.05 mol dm^{-3}) = 1/80 = **0.0125 s^{-1}** (1.25 × 10^{-2} s^{-1})

Halving the concentration halved 1/t, so rate is proportional to the thiosulfate concentration.

Exam data often uses (1/t) × 10^3 to give tidy numbers: 1/40 = 0.025, so (1/t) × 10^3 = 25.

(ii) Rate from a gradient

- On a graph of amount (or volume, or concentration) against time, the **rate at any moment = gradient of the curve** at that moment.
- The curve gets less steep with time because reactants are used up, so collisions become less frequent.
- To find the gradient of a curve, draw a **tangent**: a straight line that just touches the curve at that point, with the same slope.
- **Initial rate** = gradient of the tangent at **t = 0**. It's the fairest way to compare experiments, because nothing has been used up yet.



Volume of gas against time. The tangent at t = 0 gives the initial rate; the tangent at t = 50 s gives the rate at that moment.

Worked example: initial rate from a tangent

1 Magnesium reacts with excess hydrochloric acid and the hydrogen is collected in a gas syringe. A tangent drawn to the volume–time curve at t = 0 passes through the origin and through the point (20 s, 36 cm^3). Find the initial rate.

2 The gradient of the tangent is the initial rate:

$$\text{rate} = \frac{\Delta V}{\Delta t} = \frac{36\ \text{cm}^3}{20\ \text{s}} = 1.8\ \text{cm}^3\ \text{s}^{-1}$$

Choose two points on the tangent that are far apart: this makes the gradient more precise.

3 To express it in mol s^{-1}, use the molar volume (24.0 dm^3 mol^{-1} = 24 000 cm^3 mol^{-1} at room temperature and pressure):

$$\frac{1.8}{24\,000} = 7.5 \times 10^{-5}\ \text{mol s}^{-1}$$

WORKED EXAMPLE

Worked example: initial rate and rate at t = 50 s

Initial rate (red tangent): it passes through (0, 0) and (50, 80).

Gradient = (80 − 0) / (50 − 0) = **1.6 cm³ s⁻¹**

Rate at 50 s (green tangent): read two points far apart on the tangent, (0, 21) and (120, 92).

Gradient = (92 − 21) / (120 − 0) = 71 / 120 = **0.59 cm³ s⁻¹**

The rate has fallen to about a third of its starting value because the acid is being used up.

EXAM TIP

Draw the tangent **long**: at least half the width of the grid. Read two points **far apart** that sit on grid lines. A short tangent magnifies any reading error. Always write the units: they come from the axes (cm³ ÷ s → cm³ s⁻¹).

MARK-SCHEME WORDING

- Tangent at t = 0 must **touch** the curve and be long enough (in Jun 2022 it had to reach at least 20 cm³). Gradient accepted within a range (0.25–0.33 there). The unit **cm³ s⁻¹** was a separate mark.
- Graph marks: sensible scale using at least half the grid; **axes labelled with units**; points within ½ small square; **smooth curve** of best fit (not dot-to-dot).
- Halving the acid concentration (acid limiting): "initial rate halves **and** final volume of gas halves".

Often asked as: calculate 1/t for a table (to 3 s.f.) and plot it; draw a tangent and work out the initial rate; predict how initial rate and final volume change if the concentration or amount is changed.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- Choosing the amount of reactant so the gas fits a **100 cm³ gas syringe**, using **24 dm³ mol⁻¹** for the molar volume at room temperature (WCH12 Jun 2022).
- If the acid is the limiting reactant, halving its concentration halves the **final volume** of gas as well as the rate. If the acid is in excess, the final volume doesn't change (WCH12 Jun 2022).
- Values of 1/t in a table rounded to **3 significant figures** (WCH12 Jan 2025).

COMMON MISTAKE

No tangent drawn: the gradient was worked out from the curve's end points (that gives an *average* rate, not the rate at an instant).

Saying the final volume of gas stays the same after halving a **limiting** reactant: fewer moles of acid means less gas.

Missing units on axes, or joining points dot-to-dot.

SPEC 9.4 Maxwell–Boltzmann distribution and temperature

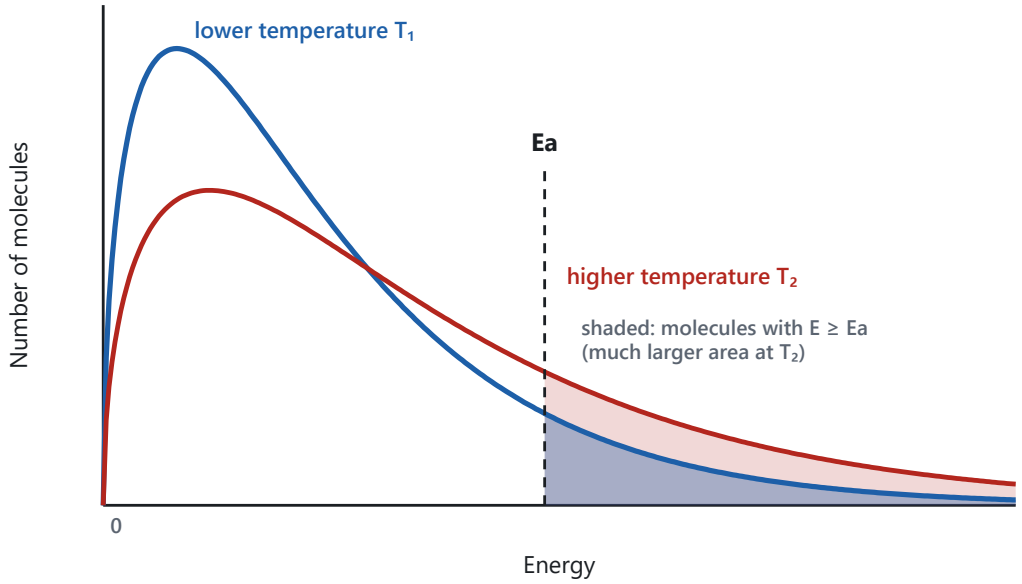
WHAT THIS SPEC POINT ASKS

Use the Maxwell–Boltzmann distribution of molecular energies to explain, qualitatively, how temperature changes the rate of reaction.

What the curve shows

- In a gas (or solution) at one temperature, the molecules don't all have the same energy. They collide and swap energy all the time, so there's a **spread** of energies.
- The **Maxwell–Boltzmann distribution** plots **number (or fraction) of molecules** (y-axis) against **energy** (x-axis).
- Features you must draw correctly:
  - starts at the **origin**: no molecules have zero energy;
  - rises to a **peak**: the most likely energy;
  - has a long tail to the right that gets closer and closer to the x-axis but **never touches it** (an **asymptote**): a few molecules have very high energy;
  - the **area under the curve** = the total number of molecules.
- Mark **E_a** to the **right of the peak**. The area under the curve to the right of E_a is the number of molecules with **enough energy to react**.

Raising the temperature



At the higher temperature T_2 the peak is lower and further right, and the area beyond E_a is much larger. The number of molecules (total area) is the same.

- The curve **flattens and shifts right**: the peak moves to a **higher energy** and becomes **lower**.
- The peak drops because the **total area stays the same**: the number of molecules hasn't changed, so if the curve spreads out it must get lower.
- The tail at high energy sits **above** the original curve, so the area beyond E_a is much bigger.
- So: at a higher temperature **many more molecules have energy ≥ E_a → more successful collisions per unit time** → faster rate. Molecules also move faster and collide more often, but that is a much smaller effect.

MARK-SCHEME WORDING

Drawing the higher-temperature curve:

- "starts at the **origin**", "**lower maximum**", "peak **moved to the right**", "**asymptotic** and **above** the original line" at high energy.
- DO NOT AWARD if any part of the new curve to the right of the peak falls below the original line, if the curves cross more than once, or if the tail ends too high (more than about a quarter of the new peak height) (WCH12 Jun 2023, Jan 2025).
- Axes: y "number / fraction / proportion of molecules", x "energy" (kinetic energy accepted).

Explaining: "more molecules / collisions have **energy greater than (or equal to) the activation energy**" or "the **area under the curve beyond E_a** increases", and "more **successful collisions per unit time**".

Often asked as: draw a second curve on a given one at a higher (or lower) temperature, then explain the rate change; part of a 6-mark question with a catalyst (9.8).

COMMON MISTAKE

Curve not starting at the origin, or the tail touching (or crossing) the x-axis.

Drawing the hotter curve with a **higher** peak: the area must stay the same.

The two curves crossing twice, or the hot curve dropping below the cold curve in the tail.

A good diagram but no **reference to it** in the written explanation: say "the area to the right of E_a is larger".

Forgetting "per unit time" in the final step.

SPEC 9.5 Catalysts: an alternative route with lower activation energy

WHAT THIS SPEC POINT ASKS

Understand that a catalyst works by providing an alternative reaction pathway with a lower activation energy.

How a catalyst works

- A **catalyst** increases the rate of a reaction **without being used up** overall. It's chemically unchanged at the end, so it can be reused.
- It provides an **alternative route (mechanism)** for the reaction, with a **lower activation energy**.
- It does **not** give particles more energy, and doesn't change ΔH. It lets **more of the collisions that already happen** succeed, because more molecules now have energy ≥ the new, lower E_a (9.8).
- The catalyst **does** take part in the reaction: it forms an **intermediate** with a reactant and is then regenerated (9.6). "Takes no part in the reaction" is wrong.
- A catalyst speeds up the forward and backward reactions equally, so it does **not** change the position of equilibrium or the yield; it just gets you to equilibrium faster (9.10).

Two kinds

	Homogeneous	Heterogeneous
Phase	same phase as the reactants	different phase (usually a solid with gas or liquid reactants)
How	forms an intermediate in solution	reactants **adsorb** onto the surface, bonds weaken and react, products **desorb**
Examples	H^+ in esterification; Fe^{2+}/Fe^{3+} in the iodide–peroxodisulfate	**iron** in the Haber process; **V_2O_5** in the Contact process; **Pt/Rh/Pd** in catalytic

Homogeneous	Heterogeneous
reaction	converters

- Heterogeneous catalysts are used as **powders, pellets or fine meshes** on a support to give a **large surface area**, so there are more sites where reaction can happen.
- They can be **poisoned**: an impurity (e.g. lead or sulfur) bonds strongly to the surface and blocks it.

MARK-SCHEME WORDING

- "A catalyst **decreases the activation energy** by providing an **alternative pathway / route / mechanism**" (WCH12 Oct 2025, Jun 2025).
- Then link it to rate: "more molecules have energy ≥ E_a" and "frequency of successful collisions increases".
- IGNORE "not used up" on its own when the question asks how it works; it's not the mechanism.

Often asked as: explain how a catalyst increases the rate; say why a substance in a process is (or isn't) a catalyst.

COMMON MISTAKE

"Catalysts do not take part in the reaction" (WCH12 Jan 2022 report). They do, and are regenerated.

"A catalyst gives the particles more energy" or "increases the yield". Neither is true.

Just writing "lowers E_a" with no mention of an **alternative route**, or no link to successful collisions.

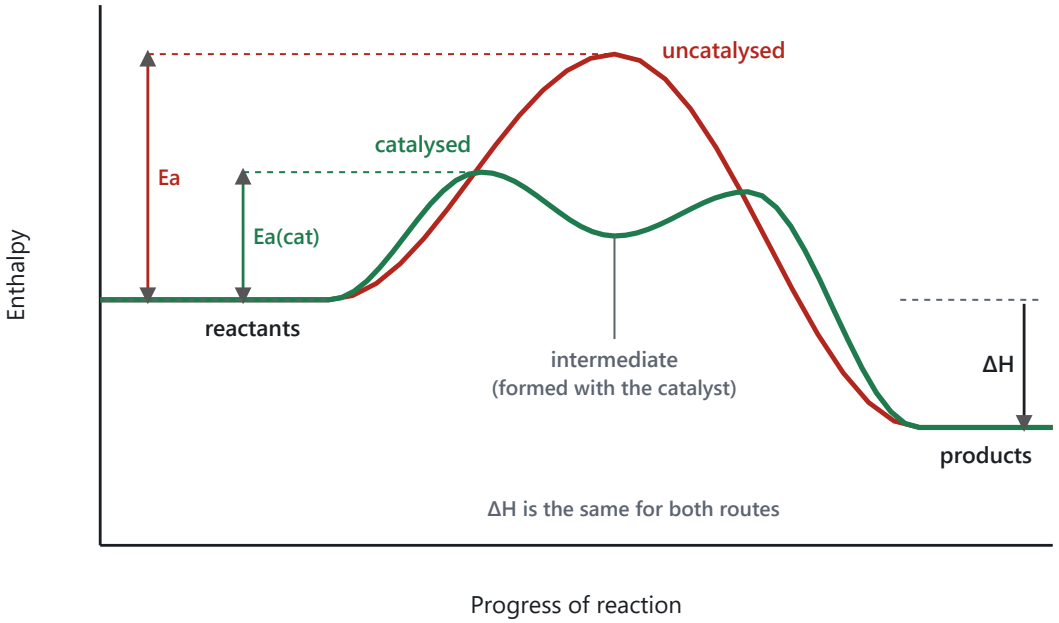
In Jan 2022 the reactant (urea) in a catalytic converter was called a catalyst: it is used up, so it isn't.

SPEC 9.6 Reaction profiles: uncatalysed and catalysed

WHAT THIS SPEC POINT ASKS

Draw reaction profiles for a reaction with and without a catalyst, including the energy level of the intermediate formed with the catalyst.

Drawing the two profiles



Exothermic reaction. Uncatalysed: one high hump. Catalysed: two lower humps with a dip between them, where the intermediate forms.

- **Same start, same finish**: the reactants and products are at the same levels on both profiles, so **ΔH is unchanged**.
- **Uncatalysed**: a single high peak. E_a = reactants → top of the peak.
- **Catalysed**: the route goes through an **intermediate** (a species made from the reactant and the catalyst). This shows as a **dip (trough)** between **two smaller peaks**:
  - step 1: reactant + catalyst → intermediate (first peak);
  - step 2: intermediate → product + catalyst (second peak), which regenerates the catalyst.
- E_a(cat) is measured from the **reactants to the highest point** on the catalysed route. Both catalysed peaks must be **lower** than the uncatalysed peak.
- The intermediate's dip sits **above the reactants** in most sketches, because the intermediate is a short-lived, fairly high-energy species. Draw it clearly as a minimum between the two humps.

WORKED EXAMPLE

Worked example: a catalysed step-by-step route

Fe^{2+} catalyses $S_2O_8^{2-} + 2I^- \rightarrow 2SO_4^{2-} + I_2$ (a reaction between two negative ions, which repel, so the uncatalysed E_a is high).

Step 1: $S_2O_8^{2-} + 2Fe^{2+} \rightarrow 2SO_4^{2-} + 2Fe^{3+}$

Step 2: $2Fe^{3+} + 2I^- \rightarrow 2Fe^{2+} + I_2$

Fe^{3+} is the **intermediate** (the dip on the profile); Fe^{2+} is regenerated, so it's the catalyst. Each step involves oppositely charged ions, so each has a lower E_a.

MARK-SCHEME WORDING

- Reactant and product levels **unchanged**; catalysed peak(s) **lower** than the uncatalysed one; E_a labelled from the **reactant line** to the peak.
- The research found no WCH12 question (2019–2026) that asked for the intermediate dip, but the spec names it, so learn it.

Often asked as: sketch a profile for a given (often endothermic) reaction and label E_a and ΔH; add a catalysed route.

COMMON MISTAKE

Moving the reactant or product level for the catalysed route (ΔH would change: it can't).

Drawing the catalysed route as one hump when the question asks for the intermediate.

Labelling E_a(cat) only to the first peak when the second peak is higher.

SPEC 9.7 Catalysts in industry: sustainability

WHAT THIS SPEC POINT ASKS

Understand how catalysts make industrial processes more sustainable by cutting energy use and/or raising atom economy.

Why industry relies on catalysts

- **Lower temperature (and sometimes pressure)** for the same rate → **less energy** needed for heating and compressing → **less fossil fuel burned**, lower CO_2 emissions and lower cost.
- **More selective (specific)** catalysts speed up only the reaction you want → **fewer side products** → **higher atom economy** and less waste to separate or dispose of.
- Allow a **different route** with fewer steps or better atom economy (e.g. an addition reaction, atom economy 100%, instead of a substitution that makes a by-product).
- They are **not used up**, so a small amount can be reused many times (though they need replacing when poisoned or worn).
- For an **exothermic** equilibrium, a catalyst lets the plant run at a **lower temperature**, which also gives a **better yield** (9.10) while keeping the rate acceptable.

Process	Catalyst	What it makes
Haber process: $N_2 + 3H_2 \rightleftharpoons 2NH_3$	iron	ammonia (fertilisers)
Contact process: $2SO_2 + O_2 \rightleftharpoons 2SO_3$	vanadium(V) oxide, V_2O_5	sulfuric acid
Hydration of ethene: $C_2H_4 + H_2O \rightleftharpoons C_2H_5OH$	phosphoric acid on silica	ethanol
Methanol: $CO + 2H_2 \rightleftharpoons CH_3OH$	copper (with ZnO/Al_2O_3)	methanol
Catalytic converter: $2CO + 2NO \rightarrow 2CO_2 + N_2$	platinum / rhodium / palladium	less toxic exhaust gases
Cracking of long-chain alkanes	zeolites	smaller alkanes and alkenes

MARK-SCHEME WORDING

- "allows a **lower temperature** to be used / **less energy** is needed (for the same rate)".
- "**specific** for the product so there is less waste / fewer other products / **improves atom economy**".
- "not used up / can be used again".
- IGNORE "just increases the rate" or "lowers E_a" when the question asks about sustainability (WCH12 Oct 2020).

Often asked as: state two ways a catalyst makes a process more sustainable (copper in methanol manufacture, phosphoric acid in ethanol manufacture).

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- Catalysts are used as **pellets or powder** for a larger surface area (WCH12 Oct 2021).
- Too much steam in ethene hydration can **wash the phosphoric acid catalyst off its support** (WCH12 Jun 2024).

COMMON MISTAKE

Answering a sustainability question with "it speeds up the reaction": link it to **energy** or **atom economy**.

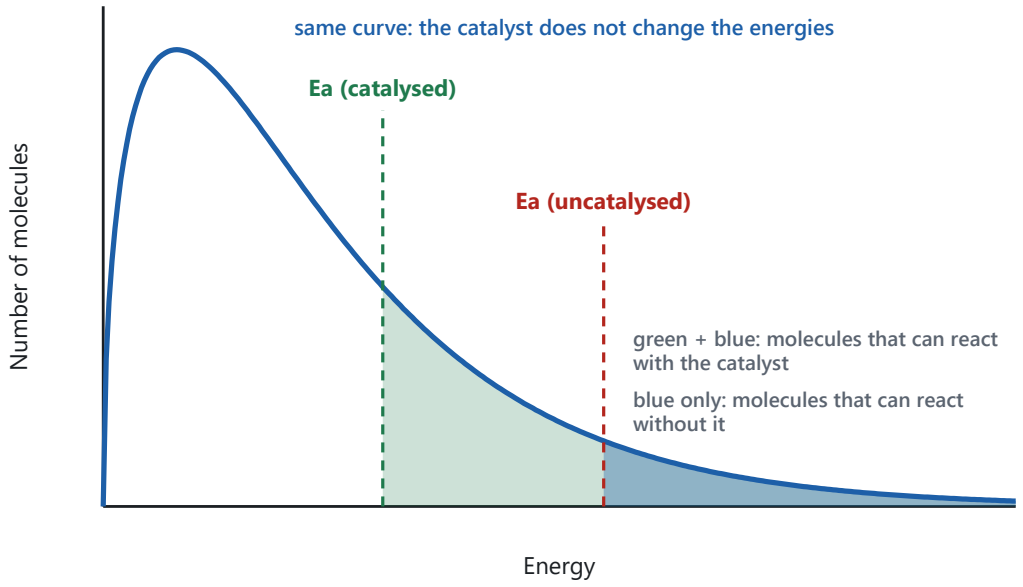
Saying a catalyst "increases the yield". For an exothermic equilibrium, it can *allow* a lower temperature that gives a better yield: say it that way.

SPEC 9.8 Catalysts and the Maxwell–Boltzmann distribution

WHAT THIS SPEC POINT ASKS

Use the Maxwell–Boltzmann distribution to explain qualitatively how a catalyst increases the rate.

One curve, two activation energies



The catalyst doesn't change the curve. It moves E_a to the left, so a much larger area lies beyond it.

- The temperature hasn't changed, so the **distribution curve stays exactly the same**. Don't draw a second curve.
- Mark the **uncatalysed E_a** and, to its **left**, the **catalysed E_a**. Both must be to the **right of the peak**.
- The area under the curve beyond E_a(cat) is much bigger than the area beyond E_a, so **many more molecules have enough energy to react → more successful collisions per unit time** → faster rate.

Temperature v catalyst: a comparison

	Raise the temperature	Add a catalyst
Curve	changes: lower peak, shifted right	unchanged
E_a	unchanged	lowered (new line to the left)
Molecules with $E \geq E_a$	more	more

	Raise the temperature	Add a catalyst
Collision frequency	increases a little	unchanged

MARK-SCHEME WORDING

Four-mark answer (WCH12 Oct 2021):

- both **axes labelled**;
- **E_a labelled**;
- catalysed E_a drawn **to the left** of the uncatalysed E_a (both to the right of the peak, on the same curve);
- "**more particles have energy greater than E_a**" / "greater area to the right of E_a(cat)".

Then for rate: "increases the number of successful collisions **per unit time**". Shading isn't required, but wrong shading loses the mark.

DO NOT AWARD "catalyst increases yield", "particles have more energy", "move faster" or "greater collision frequency".

Often asked as: a 6-mark question with both temperature and a catalyst on one diagram (WCH12 Oct 2019); draw catalysed E_a on a given curve and explain.

COMMON MISTAKE

Drawing a new curve for the catalyst.

Putting E_a(cat) at or to the left of the peak.

Losing the last mark by not saying **per unit time** or **frequency** (WCH12 Jan 2022 report).

SPEC 9.9 Reversible reactions and dynamic equilibrium

WHAT THIS SPEC POINT ASKS

Know that many reactions are reversible and can reach a dynamic equilibrium in which the forward and backward rates are equal and the concentrations of reactants and products stay constant.

Reversible reactions

- A reversible reaction can go both ways: reactants → products **and** products → reactants. The sign is ⇌.
- Example: $N_2O_4(g) \rightleftharpoons 2NO_2(g)$ (colourless ⇌ brown).

- Example: $CuSO_4 \cdot 5H_2O(s) \rightleftharpoons CuSO_4(s) + 5H_2O(l)$ (blue crystals ⇌ white powder; heat drives it right, adding water drives it left).

Dynamic equilibrium

Start with only reactants:

- At first the forward rate is high (lots of reactant) and the backward rate is zero (no product).
- As reactants are used up the forward rate falls; as products build up the backward rate rises.
- Eventually the two rates become **equal**. From then on, each reaction makes particles as fast as the other uses them up.

At **dynamic equilibrium**:

- the **rate of the forward reaction equals the rate of the backward reaction**;
- the **concentrations of reactants and products stay constant** (they are **not** usually equal).

- "Dynamic" because both reactions **are still happening**; at the particle level nothing has stopped.
- On the large scale nothing appears to change: the **colour, pressure, concentration** stay constant.
- It needs a **closed system** if any substance is a gas, so nothing can escape. (An open flask of solution can reach equilibrium; heating $CaCO_3$ in an open tube can't, because CO_2 escapes and the reaction goes to completion.)
- The same equilibrium can be reached **from either direction**: starting from products gives the same equilibrium mixture at the same conditions.

Suggested practicals

- **NO_2 / N_2O_4**: $2NO_2(g) \rightleftharpoons N_2O_4(g)$, $\Delta H = -57$ kJ mol^{-1}. NO_2 is **brown**, N_2O_4 is **colourless**. Sealed tubes of the mixture go **darker in hot water** and **paler in ice**; squeezing a syringe of the gas (higher pressure) makes the mixture paler once it settles (see 9.10).
- **ICl / ICl_3**: $ICl(l) + Cl_2(g) \rightleftharpoons ICl_3(s)$. ICl is a **brown liquid**, ICl_3 is a **yellow solid**. Passing more Cl_2 in gives more yellow solid; letting Cl_2 out (turning the U-tube upside down) gives back the brown liquid.

MARK-SCHEME WORDING

- Equation questions: "products **and reversible arrow**" (ALLOW two full-headed arrows) (WCH12 Jan 2022).
- Definitions: "rate of forward reaction = rate of backward reaction" and "concentrations (of reactants and products) **remain constant**".

Often asked as: rarely as a bare definition. It hides inside industrial questions, where you must recognise that a reaction is an equilibrium and can't go to completion.

COMMON MISTAKE

"At equilibrium the concentrations of reactants and products are **equal**." They're **constant**, not equal.

"The reaction has stopped." Both reactions continue at the same rate.

Treating an equilibrium as if it goes to completion, e.g. "excess steam will react with all the methane" (WCH12 Jan 2023 report).

SPEC 9.10 Position of equilibrium: temperature, pressure and concentration

WHAT THIS SPEC POINT ASKS

Predict and justify how changing temperature, pressure and concentration shifts the position of a homogeneous equilibrium.

Le Chatelier's principle

- If a system at equilibrium is disturbed, the **position of equilibrium shifts to oppose the change**.
- "Position shifts **right**" = more products (higher yield). "Shifts **left**" = more reactants.
- **Homogeneous** = everything in the same phase (e.g. all gases, or all in solution).

The three changes

Change	Equilibrium shifts…	Why (opposes the change)
Increase concentration of a reactant	right	uses up some of the added reactant
Remove a product as it forms	right	replaces the product
Increase pressure (gases)	to the side with **fewer moles of gas**	fewer gas molecules → lowers the pressure
Decrease pressure	to the side with more moles of gas	more molecules → raises the pressure
Increase temperature	in the **endothermic** direction	absorbs the extra heat
Decrease temperature	in the **exothermic** direction	releases heat

Change	Equilibrium shifts…	Why (opposes the change)
Add a catalyst	**no shift**	speeds up both reactions equally; equilibrium is reached sooner

- Pressure: count **gas** moles only (ignore solids and liquids). If both sides have the same number of gas moles, pressure has **no effect** on the position.
- Temperature: use the **sign of ΔH** for the **forward** reaction. ΔH negative → forward is exothermic → a **lower** temperature shifts it right.
- Only temperature changes the value of the equilibrium constant, K (Unit 4). Concentration and pressure move the position but leave K unchanged.

WORKED EXAMPLE

Worked example: three changes on one equilibrium

$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$ $\Delta H = -197$ kJ mol^{-1}

Higher pressure: left side has 3 moles of gas, right side has 2. The equilibrium shifts **right** (fewer gas moles), so the yield of SO_3 **increases**.

Higher temperature: forward reaction is **exothermic**, so the endothermic (backward) direction is favoured. Equilibrium shifts **left**; yield **decreases**.

Adding more O_2: equilibrium shifts **right** to use up the extra O_2; more SO_3.

WORKED EXAMPLE

Worked example: a colour change in solution

$2CrO_4^{2-}(aq) + 2H^+(aq) \rightleftharpoons Cr_2O_7^{2-}(aq) + H_2O(l)$

yellow ⇌ orange

Add acid (more H^+): shifts **right** → the solution turns **orange**.

Add alkali (OH^- removes H^+): shifts **left** → turns **yellow**.

EXAM TIP

For a 2–3 mark "predict and justify" answer write: **direction of shift** + **the reason** (number of gas moles on each side, or sign of ΔH) + **effect on yield**. If you quote mole numbers they must be right.

MARK-SCHEME WORDING

- Pressure: "higher pressure moves the position of equilibrium **to the right / forward** / increases the yield" and "**4 gaseous moles on the left and 2 on the right**" (Haber, WCH12 Oct 2020). A wrong direction scores 0. Quoted mole numbers must be correct.
- Temperature: "the (forward) reaction is **exothermic**", "a **lower temperature** would increase the yield / move the equilibrium to the right". If you call the forward reaction endothermic, neither mark is scored.
- Removing product: "remove the product (as it is formed)", e.g. "cool so ethanol (and water) condense".
- Adding NaOH to chlorine water moves its equilibrium right because the NaOH **removes H^+ / HCl** (WCH12 Oct 2024).

Often asked as: effect of pressure or temperature on the yield of an industrial process; why a product is removed; inside a 6-mark compromise question (9.11).

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- More steam pushes ethene hydration to the right but **dilutes** the product and can wash the catalyst off its support (WCH12 Jun 2024).
- Removing CO_2 from a hydrogen-making process shifts the equilibrium towards more H_2 (WCH12 Jan 2023).

COMMON MISTAKE

"Higher temperature increases the yield because it increases the rate" (WCH12 Jan 2021 report). Rate and yield are different questions.

Counting all moles instead of **gas** moles, or saying "atoms".

Writing "shifts left" and then giving a reason that supports a shift right (WCH12 Oct 2022 report).

Assuming a reaction is endothermic just because it's heated (WCH12 Jan 2022 report): check the sign of ΔH.

Saying a catalyst moves the equilibrium.

SPEC 9.11 Industrial compromise: yield v rate

WHAT THIS SPEC POINT ASKS

Use data to explain why many industrial processes must compromise between a high yield and a fast rate.

Why a compromise is needed

For an **exothermic** reaction with **fewer gas moles on the product side** (very common):

Condition	Effect on **yield** (equilibrium)	Effect on **rate** (kinetics)	Other costs
High temperature	lower: favours the endothermic backward reaction	faster: more molecules with E ≥ E_a	more energy (fuel) to heat
Low temperature	higher	**too slow**	
High pressure	higher: shifts to fewer gas moles	faster: more collisions per unit time	expensive thick-walled plant and pumps, more energy to compress, safety risk
Catalyst	no effect	faster at a lower temperature	cost of catalyst, poisoning

- So a **compromise temperature** is chosen: low enough for an acceptable yield, high enough for an acceptable rate.
- A **compromise pressure** is chosen: high enough for a good yield and rate, but not so high that the plant is too expensive or dangerous.
- A **lower yield** each time through is acceptable because **unreacted gases are recycled**, so almost all the starting material is converted eventually. Removing the product as it forms also pulls the equilibrium right.
- Making product **fast** at a modest yield often gives more product **per day** than a high yield made very slowly.

Process	Temperature	Pressure	Catalyst	Yield per pass
Haber: $N_2 + 3H_2 \rightleftharpoons 2NH_3$, $\Delta H = -92$ kJ mol^{-1}	about 450 °C	about 200 atm	iron	about 15% (unreacted gas recycled)
Hydration of ethene: $C_2H_4 + H_2O \rightleftharpoons C_2H_5OH$, $\Delta H = -45$ kJ mol^{-1}	about 300 °C	about 60–70 atm	phosphoric acid on silica	about 5% (ethene recycled)
Contact: $2SO_2 + O_2 \rightleftharpoons 2SO_3$, $\Delta H = -197$ kJ mol^{-1}	about 450 °C	1–2 atm	V_2O_5	over 99% (so no high pressure needed)

WORKED EXAMPLE

Worked example: model answer for a 6-mark compromise question

Methanol: $CO(g) + 2H_2(g) \rightleftharpoons CH_3OH(g)$, $\Delta H = -91$ kJ mol^{-1}. Conditions used: 250 °C, 50–100 atm, copper catalyst. Explain the choice.

- **Temperature (rate)**: a high temperature gives a faster rate because more molecules have energy ≥ E_a, so there are more successful collisions per unit time.
- **Temperature (yield)**: the forward reaction is exothermic, so a high temperature moves the equilibrium left and lowers the yield. 250 °C is a **compromise** between rate and yield.
- **Pressure (yield)**: 3 moles of gas on the left, 1 on the right; high pressure moves the equilibrium to the right (fewer gas moles), increasing the yield. It also increases the rate.
- **Pressure (cost)**: very high pressure needs expensive, thick-walled equipment and a lot of energy for compressors, so a moderate pressure is used.
- **Catalyst**: copper provides an alternative route with lower E_a, so the rate is acceptable at the lower temperature; it doesn't affect the yield.
- **Recycling**: unreacted CO and H_2 are separated and recycled, so a moderate yield per pass is acceptable.

MARK-SCHEME WORDING

Indicative points (WCH12 Jan 2021, Oct 2025, Oct 2021):

- "high temperature increases the rate (more particles with E ≥ E_a)";
- "a catalyst increases the rate by providing an alternative mechanism with lower E_a";
- "high temperature reduces yield / moves equilibrium left as the reaction is **exothermic**";
- "so the temperature is a **compromise**";
- "high pressure increases yield (equilibrium moves to the side of fewer gas molecules) **or** rate";
- "high pressure: **equipment cost**", "high temperature: **energy cost**";
- "low yield is acceptable because **unreacted reactants are recycled**" (ALLOW "product removed as it forms").

IGNORE environmental effects and atom economy in these questions.

Often asked as: a 6-mark question on a named process (ethanol, methanol, hydrogen) giving the conditions and ΔH; sometimes with a yield-against-pressure graph at several temperatures to evaluate.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- **Recycling** unreacted gases and **removing the product** (e.g. cooling so ethanol condenses) (WCH12 Jan 2021, Jun 2024).

- Engineering costs: **thicker pipe walls**, compressor energy, safety of high-pressure plant (WCH12 Oct 2025, Jun 2024).
- A higher-temperature route may be chosen because it makes a **useful by-product** (e.g. H_2) (WCH12 Jun 2023).

COMMON MISTAKE

Covering only temperature: unstructured answers miss a variable. Take **temperature, pressure, catalyst, recycling** in turn (WCH12 Jan 2021 report).

Explaining the cost of high pressure without saying **why** it costs more (thicker equipment, energy for compressors) (WCH12 Jan 2025 report).

Justifying a 5% yield because "a lot of product is made": the reason is **recycling**.

Quick checks

QUICK CHECK

What does the total area under a Maxwell–Boltzmann distribution curve represent?

- The activation energy
- The total number of molecules
- The mean energy of the molecules
- The number of molecules that can react

The curve shows how many molecules have each energy, so adding them all up (the area) gives the total number of molecules. Only the area beyond E_a is the number that can react.

QUICK CHECK

Which statement best explains why a small increase in temperature greatly increases the rate of a reaction?

- The activation energy decreases.
- The particles collide more often.
- A much greater proportion of the colliding particles have energy at least equal to the activation energy.
- The enthalpy change of the reaction becomes more negative.

The tail of the Maxwell–Boltzmann curve beyond E_a grows a lot when the temperature rises. More frequent collisions also help, but that effect is small. The activation energy itself does not change with temperature.

QUICK CHECK

How does a catalyst increase the rate of a reaction?

- It gives the reacting particles more kinetic energy.
- It increases the frequency of collisions between particles.
- It makes the reaction more exothermic.
- It provides an alternative route with a lower activation energy.

With a lower E_a, more of the existing molecules have enough energy to react. The distribution of energies and ΔH are unchanged.

QUICK CHECK

For $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$, $\Delta H = -92$ kJ mol^{-1}, what happens when the temperature is increased?

- The rate increases and the equilibrium yield of ammonia decreases.
- The rate increases and the equilibrium yield of ammonia increases.
- The rate decreases and the equilibrium yield of ammonia increases.
- The rate increases and the equilibrium yield of ammonia stays the same.

Higher temperature always increases the rate. The equilibrium shifts in the endothermic direction, which is backwards here because the forward reaction is exothermic, so less ammonia is present at equilibrium.

QUICK CHECK

The pressure on an equilibrium mixture of $H_2(g)$, $I_2(g)$ and HI(g) is doubled by halving its volume. What do you observe?

- The purple colour fades as the equilibrium shifts to the right.
- The purple colour fades as the equilibrium shifts to the left.
- The purple colour darkens, and the position of equilibrium does not change.
- There is no change in colour.

There are two gas molecules on each side, so pressure does not move the position of equilibrium. But the same amount of iodine is now in half the volume, so its concentration doubles and the colour darkens.

QUICK CHECK

Which statement about a system in dynamic equilibrium is correct?

- The concentrations of the reactants and products are equal.
- The forward and backward reactions continue at equal rates.
- Both the forward and backward reactions have stopped.
- Adding a catalyst moves the position of equilibrium to the right.

At dynamic equilibrium both reactions keep going at the same rate, so the concentrations stay constant, but they are not usually equal. A catalyst speeds up both reactions equally and does not move the position of equilibrium.

QUICK CHECK

Why is the Contact process run at only 1–2 atm even though higher pressure would increase the yield of SO_3?

- Higher pressure would slow the reaction down.
- The catalyst only works at low pressure.
- Higher pressure would shift the equilibrium to the left.
- The yield is already very high, so the extra cost of high pressure is not worth it.

$2SO_2 + O_2 \rightleftharpoons 2SO_3$ has fewer gas molecules on the right, so higher pressure would increase the yield, but the equilibrium already lies far to the right at moderate pressure. Compressors and strong equipment would cost more than the small gain is worth.